



Structure of a prokaryotic SEFIR domain reveals two novel SEFIR-SEFIR interaction modes

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ABSTRACT

SEFIR domain-containing proteins are crucial for mammalian adaptive immunity. As a unique intracellular signaling domain, the SEFIR-SEFIR interactions mediate physical protein-protein interactions in the immune signaling network, especially the IL-17- and IL-25-mediated pathways. However, due to the lack of structural information, the detailed molecular mechanism for SEFIR-SEFIR assembly remains unclear. In the present study, we solved the crystal structures of a prokaryotic SEFIR domain from *Bacillus cereus* F65185 (BcSEFIR), where the SEFIR domain is located at the N terminus. The structure of BcSEFIR revealed two radically distinct SEFIR-SEFIR interaction modes. In the asymmetric form, the C-terminal tail of one SEFIR binds to the helix αA and βB - $\alpha B'$ segment of the other one, while in the symmetric form, the helices ηC and αE and the DE-segment compose the inter-protomer interface. The C-terminal tail of BcSEFIR, critical for asymmetric interaction, is highly conserved among the SEFIR domains of Act1 orthologs from different species, in particular three absolutely conserved residues that constitute an EXXXPP motif. In the symmetric interaction mode, the most significant contacts made by residues on helix αE are highly conserved in Act1 SEFIR domains, constituted an RLI/LXE motif. The two novel SEFIR-SEFIR interaction modes might explain the structural basis for SEFIR domain-mediated complex assembly in signaling pathways.

1. Introduction

SEFIR [SEF (similar expression to fibroblast growth factor genes) and IL-17R] domain-containing proteins, including interleukin-17 receptor (IL-17R) family (IL-17RA to IL-17RE) and their adaptor protein NF- κB activator 1 [Act1; also known as connection to I- κB kinase and stress-activated protein kinase (CIKS)], are crucial for mammalian adaptive immunity (Novatchkova et al., 2003; Sonder et al., 2011; Pisitkun et al., 2010; Sonder et al., 2012; Li et al., 2000; Moynagh, 2005). These SEFIR domain-containing proteins have been shown to be associated with many autoimmune diseases in both humans and mice (Sonder et al., 2011; Pisitkun et al., 2010; Chang et al., 2006; Kang et al., 2013; Giltaiy et al., 2013; Ryzhakov et al., 2012; DeSelm et al., 2012). Upon stimulation, SEFIR domain-containing receptors, such as IL-17R (IL-17RA and IL-17RC) and IL-25R (IL-17RA and IL-17RB), recruit the cytoplasmic adaptor protein Act1 via heterotypic SEFIR-SEFIR interaction (Chang et al., 2006; Estefania Claudio et al., 2009; Gaffen, 2009; Fitzgerald et al., 2001). Act1 can also form oligomers via

homotypic SEFIR-SEFIR interaction (Mauro et al., 2003; Ryzhakov et al., 2011; Zhang et al., 2013). Therefore, SEFIR domain is well known as a functional signaling domain to mediate protein-protein interaction.

SEFIR domain shares limited sequence homology with Toll/interleukin-1 receptor/resistance (TIR) domain, which has also been proved to mediate protein-protein interactions in various signal transduction pathways (Fitzgerald et al., 2001; Iwasaki and Medzhitov, 2004; Watters et al., 2007; von Bernuth et al., 2008; Dangl et al., 2013; Medzhitov et al., 1997). SEFIR and TIR domain-containing proteins together constitute a large superfamily named STIR (SEFIR/TIR) (Novatchkova et al., 2003). Of all the STIRs, many TIR domains structures, including mammalian, plant and bacterial TIR domains have been determined (Ve et al., 2015; Halabi et al., 2017; Zhang et al., 2017; Williams et al., 2016; Enokizono et al., 2013; Hyun et al., 2016; Nishimura et al., 2017), but only two monomer structures of SEFIR domain (human IL-17RA-SEFIR and mouse IL-17RB-SEFIR) were reported (Zhang et al., 2013, 2014). These two SEFIR structures share similar α/β fold with TIR domains, and display canonical STIR folding

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as a compact globular architecture comprised of a central parallel β -sheet wrapped by several α -helices (Zhang et al., 2013, 2014). However, the monomer structures could not provide structural information for SEFIR-SEFIR assembly mode, the major function of this domain. Previous studies have suggested that SEFIR domain has specific structural elements for IL-17R signaling compared with TIR domain (Zhang et al., 2013; Hyun et al., 2016; Liu et al., 2011). Moreover, mutagenesis experiments demonstrated the distinct interfaces for Act1-SEFIR to interact with IL-17RA-SEFIR or to form oligomers (Liu et al., 2011; Boisson et al., 2013). Therefore, the detailed structural information for SEFIR-SEFIR interaction has been of great interest and importance.

The SEFIR domain-containing proteins have been found throughout the animal kingdom and even in prokaryotes (Novatchkova et al., 2003; Ryzhakov et al., 2011; Wu et al., 2012, 2011). Comparative and phylogenetic sequence analyses suggested that bacterial SEFIR homologues may be acquired through lateral gene transfer from eukaryotes (Wu et al., 2012). Prokaryotic SEFIR domain-containing proteins with various domain architectures occur in a wide range of bacteria, almost certainly play diverse roles in bacterial physiology. SEFIR domains may function as general protein-protein interaction domains for diverse physiological actions. To elucidate the detailed SEFIR-SEFIR interaction modes, we have determined crystal structures of a prokaryotic SEFIR domain, unraveling two novel SEFIR-SEFIR interaction forms (asymmetrical and symmetrical) mediated by distinct binding sites. This study provides the structural basis for SEFIR-mediated protein-protein interactions.

2. Results

2.1. Crystal structures of BcSEFIR

Multiple sequence alignment indicates that SEFIR domains from different proteins share an extended C-terminal tail region (up to 18 residues in human Act1) (Fig. 1A), which has not been explicitly described in previously reported SEFIR structures (Zhang et al., 2013, 2014). While the SEFIR domain has been implicated in protein-protein interactions, the function of this C-terminal tail remains unclear. We aimed to investigate the structure and function of this tail region in the context of SEFIR domain by using a prokaryotic SEFIR domain from *Bacillus cereus* F65185 (BcSEFIR) as a model, which is located at the N-terminus of the protein.

According to the multiple sequence alignment (Fig. 1A), two constructs containing BcSEFIR_{tail} (amino acid residues 1–143) and BcSEFIR (residues 1–153) were designed. Their structures were determined in high resolution of 2.0 Å and 1.8 Å, respectively. All the residues in the constructs could be readily traced, including those located at the loop regions. As expected, the overall structure of BcSEFIR exhibits a folding of typical α/β domains like other STIR superfamily members, which was characterized by a central parallel β -sheet flanked by several α -helices on both sides (Fig. 1B). Superposition of BcSEFIR and BcSEFIR_{tail} showed that truncation of C-terminal tail had limited influence on the conformation of the globular core of BcSEFIR (r.m.s. deviation of 0.82 Å for all atoms), except for minor conformational differences at the regions around CC-loop or DE-segment (Fig. 1C), probably due to different molecular packing patterns in two crystal forms. The C-terminal 10-residue tail of BcSEFIR does not attach to the globular α/β core or adopt a disordered state with high flexibility. Instead, the tail extended outward to form extensive interactions with an adjacent BcSEFIR molecule in the crystal.

The central β -sheet of BcSEFIR consists of five β -strands (β A– β E) and is slightly twisted. In the β -sheet of a typical TIR domain, such as TLR1-TIR and TLR2-TIR (Xu et al., 2000), each β -strand is followed by only one α -helix (α A– α E). However, BcSEFIR adopts a more complex structural configuration, where the region between two adjacent β -strands contains more secondary structural elements than a single α -helix (Fig. S1). The BC-segment located between β B and β C strands

possesses two α -helices (α B' and α B), while CD-segment is composed by two α -helices (α C' and α C) and one 3_{10} -helix (η C). The DE-segment is even more unusual, comprising no helices, but a β -hairpin formed by two short β -strands (β D' and β D'') (Fig. S1). The BB-loop and CC-loop (important for STIR-STIR interactions (Liu et al., 2011; Xu et al., 2000; Toshchakov et al., 2011, 2005; Tao et al., 2002; Williams et al., 2014)) are equivalent to the loops linking α B' and α B, and α C' and α C, respectively (Fig. 1B). These two regions are fully exposed to solvent in BcSEFIR domain. The short α B' helix in BcSEFIR adopts a stable and rigid conformation through interactions with helix α A and α B (Fig. 1D). Residue Trp-38 on helix α B' is associated with a hydrophobic platform containing residues Leu-19 and Ala-22 on helix α A. In addition to the intensive hydrophobic interactions, residue Asp-36 on helix α B' is hydrogen bonded to Tyr-7 on AA-loop (2.68 Å). Interestingly, the Tyr-7 and Asp-36 are highly conserved in Act1-SEFIR (Fig. S2), indicating that this hydrogen bond could be formed in Act1-SEFIR, tethering α B' helix to adopt a stable conformation.

2.2. Structural comparison of BcSEFIR with IL-17Rs-SEFIR

Although BcSEFIR has low amino acid sequence similarity with the SEFIR domains of IL-17RB (14.6% identity) and IL-17RA (18.3% identity), they indeed share similar overall topology. Superposition of BcSEFIR and IL-17RB-SEFIR gave an r.m.s. deviation of 2.27 Å for 107 aligned C α atoms (Fig. 2A), indicating that bacterial SEFIR domains share similar fold with mammalian SEFIR. Between determined prokaryotic and eukaryotic structures, major differences lie in the BC-, CD- and DE-segments. The CC-loop of IL-17RB-SEFIR cannot be observed in its crystal structure, due to its high flexibility. Accordingly, a model was proposed with crossed CC- and DD-loops, exhibiting an unusual knot topology (Fig. 2C) (Zhang et al., 2013). However, in the structures of BcSEFIR, all loop regions are well ordered without any knot topology formed.

The structure of BcSEFIR partially resembles that of IL-17RA-SEFIR, with an r.m.s. deviation of 1.92 Å over 119 aligned C α atoms. Although the core structural elements of these two structures share canonical STIR folding, the helices, β -sheets and loops connecting them in BcSEFIR differ from that in IL-17RA-SEFIR in their lengths and orientations. The significant differences between the two structures lie in CD-segment (α CC'_{ins}, α C', CC loop) and DE-segment (α D, DD loop) (Fig. 2B).

2.3. Asymmetric interaction mode observed in BcSEFIR structure

The crystal structure of BcSEFIR contains only one protein molecule per asymmetric unit, but we noticed an interesting interaction form between two adjacent BcSEFIR domains in the crystal. The C-terminal tail (CT) of one BcSEFIR (referred to as the tail-presenting) attaches to the cleft between the α A helix and β B– α B' segment of the adjacent BcSEFIR molecule (referred to as the tail-docking) via extensive interactions, where most of the BcSEFIR CT residues are involved (Fig. 3). Multiple sequence alignment of SEFIR domains from different species (Fig. 1A) indicates that the extended tails at the C-terminus are ubiquitous in eukaryotic ones, implicating potential biological importance of the extended C-terminal tail.

At the C-terminus of BcSEFIR, Arg-152 is the key residue mediating the anchoring of this tail portion (Fig. 3F). An ionic bond is formed between this Arg-152 and the β B Glu-32 of the tail-docking subunit. There are other two hydrogen-bonds between this Arg-152 and the Ser-0 at the very N-terminus of the tail-docking BcSEFIR. It is important to note that the Ser-0 is an artificially introduced amino acid residue, while the natural residue valine at this position might not form hydrogen-bonds via its side chain. At the N-terminus of BcSEFIR CT, the key residue is Glu-144 that forms two hydrogen-bonds with the α B' Arg-37 of the tail-docking subunit (Fig. 3E). BcSEFIR CT also exhibits hydrophobic interactions between its intermediate portion (Leu-146–Pro-

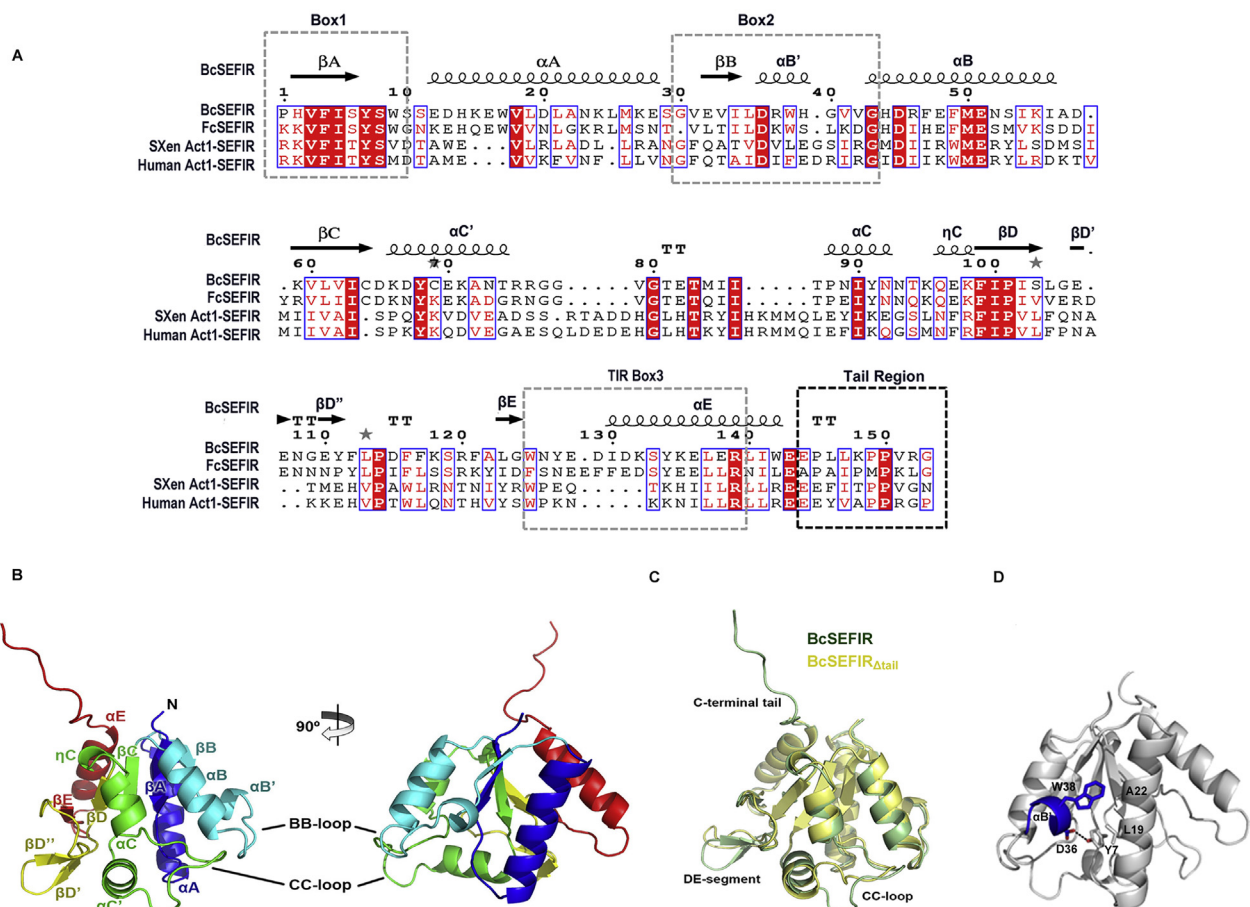


Fig. 1. Crystal structures of BcSEFIR. (A) Alignment of the SEFIRs in different species. Secondary structures shown above are according to BcSEFIR (β-sheets, black arrows; α-helices, cylinders). SEFIR domain from *Bacillus cereus*, *Flavobacterium caucensis*, *Xenopus (Silurana) tropicalis*, human are used in the alignment. (B) Overall structure of BcSEFIR depicted in cartoon representation with secondary elements colored differently and labeled appropriately. (C) Superposition of BcSEFIR and BcSEFIR_{Δtail} structures. Regions with conformational difference are labeled. (D) The short helix αB' (blue) is tethered by hydrophobic interactions and hydrogen-bond. Residue Asp-36 in helix αB' forms a hydrogen-bond with Tyr-7 from AA-loop.

150) and the small hydrophobic patch in the cleft (including Leu-19, Met-26 on αA and Trp-38 on αB') Fig. 3E and F). Because of those extensive interactions, the C-terminal tail of the tail-presenting BcSEFIR is tightly fixed at the surface of the tail-docking BcSEFIR. The BcSEFIR CT is conserved among Act1 SEFIR domains, the three absolutely conserved residues including Glu-144, Pro-149, Pro-150 constitute an EXXXPP motif (Fig. S3). Additionally, the αE and αA helices of the tail-presenting BcSEFIR interact with the αB of the tail-docking BcSEFIR, further stabilizing the asymmetric SEFIR-SEFIR association (Fig. 3B). Hydrophobic interactions play an important role at this interface, where the αE Trp-142 of the tail-presenting BcSEFIR and the αB Val-42 of the tail-docking BcSEFIR are the central residues. Notably, there is also a hydrogen-bond between the side chain nitrogen atom of αB His-44 from the tail-docking subunit and the main chain oxygen atom of αA Glu-28 from the tail-presenting subunit Fig. 3D).

2.4. Symmetric interaction mode observed in BcSEFIR_{Δtail} structure

Without the C-terminal 'tail', two BcSEFIR_{Δtail} molecules form a dimer in a symmetric manner distinguished from BcSEFIR, exhibiting a distinct dimerization interface Fig. 4). The area of the BcSEFIR_{Δtail} dimerization interface is ~958 Å², approximately 1.5-fold of the asymmetric interface of BcSEFIR (~662 Å²). The symmetric BcSEFIR_{Δtail} dimer exhibits a canonical pattern of protein-protein interactions: a hydrophobic core patch surrounded by hydrogen-bond connections. The hydrophobic core is mainly comprised by the Phe-121 at the C-terminus of βE strand and the Leu-140 of αE helix. The benzene ring of

Phe-121 is inserted into the hydrophobic patch of the opposite BcSEFIR_{Δtail} surface, forming a solid association Fig. 4. Several hydrogen-bonds are built between the αE helix of one BcSEFIR_{Δtail} (residues Lys-135, Arg-139 and Glu-143) and the DE-segment of the opposite one (residues Asn-108 and Arg-120), as well as between the two ηC helices from each side (residues Gln-96 and Glu-97). The interacting residues Arg-139, Leu-140, Glu-143 are highly conserved in Act1 SEFIR domains, constituted an RLI/LXE motif Fig. 1A).

Taken together, the characterization of two distinct SEFIR-SEFIR interaction forms (asymmetrical and symmetrical) illustrates two plausible interaction modes via distinct binding sites on SEFIR domains Fig. 5A and B). In addition, the symmetric interface found in BcSEFIR_{Δtail} cannot be observed in BcSEFIR crystal lattice. And there is no shared interface in the two structures. Several other SEFIR-SEFIR domain interfaces have also been observed in the crystals of BcSEFIR and BcSEFIR_{Δtail}. However, most are crystallization contacts with much smaller interfaces as well as less interacting residues (Table S1), which may not represent physiological interaction.

To quantitatively investigate the BcSEFIR-BcSEFIR interaction in solution, we performed Surface Plasmon Resonance (SPR) experiment. The binding affinities between BcSEFIR-BcSEFIR, BcSEFIR_{Δtail}-BcSEFIR_{Δtail}, BcSEFIR-BcSEFIR_{Δtail} were all assessed (Fig. S5). These SPR results were consistent with the SEFIR-SEFIR interaction modes we observed in crystals, that BcSEFIR domain has different binding sites, and the tail region does not disturb the binding of the other interface. It has been reported that most isolated recombinant mammalian STIR

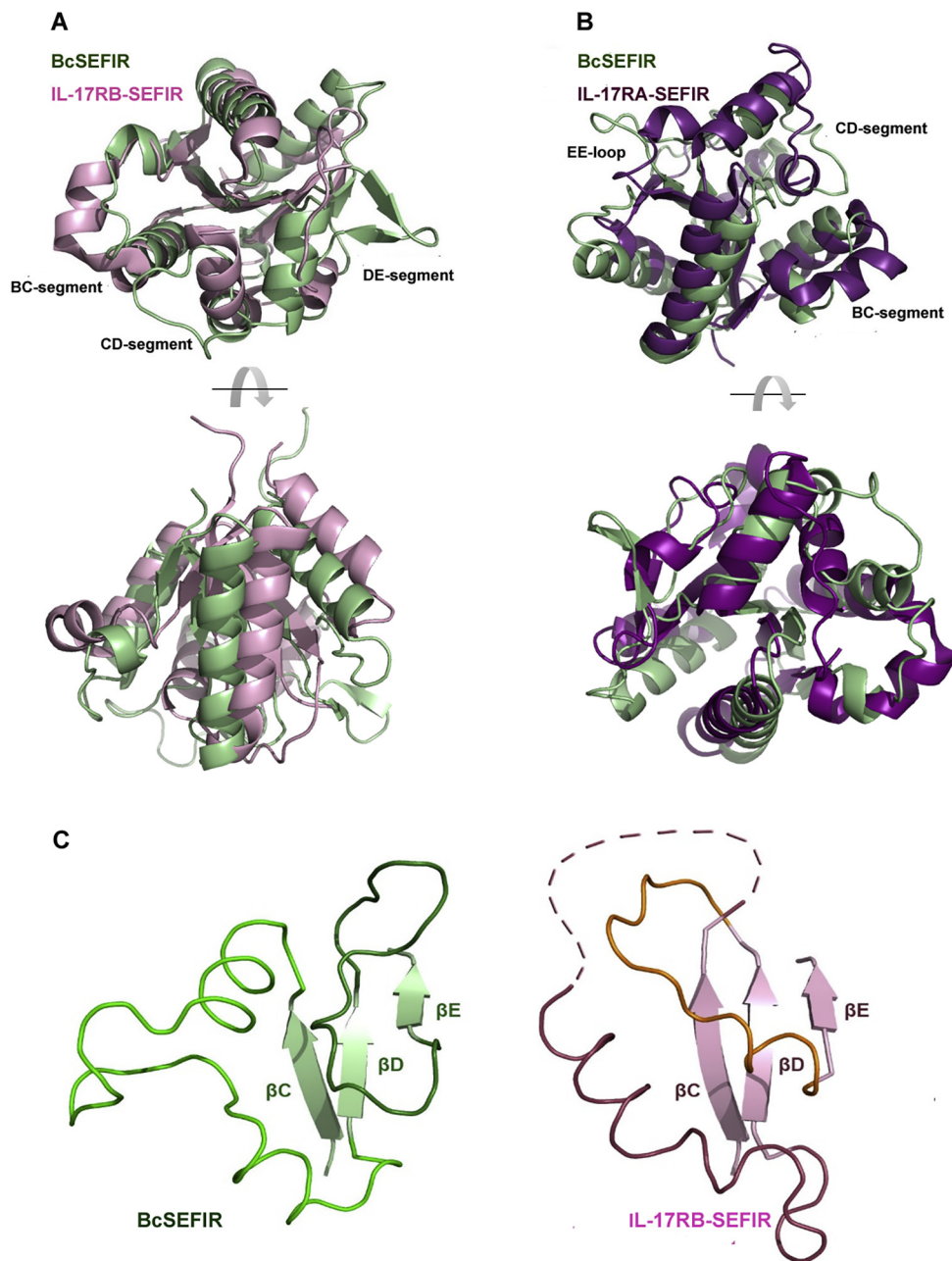


Fig. 2. Superposition of BcSEFIR with other SEFIR domain structures. (A) Superposition of BcSEFIR and IL-17RB SEFIR domain (PDB code: 3VBC). (B) Superposition of BcSEFIR and IL-17RA SEFIR domain (PDB code: 4NUX). (C) The segment between β C and β D is entirely ordered in BcSEFIR structure and does not form the unusual knot topology as suggested by the structure of IL-17RB-SEFIR.

domains formed monomers in solution, such as IL-17RA-, IL-17RB SEFIR and TIR domain-containing adaptor proteins (Zhang et al., 2013, 2014; Valkov et al., 2011; Lin et al., 2012; Nyman et al., 2008; Alaidarous et al., 2014). In our binding assay, BcSEFIR domain also showed weak affinity for self-association in solution. Moreover, in the gel filtration chromatography, BcSEFIR was eluted as monomer in solution (Fig. S6). These results suggested that weak interaction and transient oligomers of STIR domains may be a general feature of this superfamily.

2.5. Validation of key residues for BcSEFIR-SEFIR interactions

To validate the significance of the two SEFIR-SEFIR interaction modes observed in the crystals, mutagenesis experiments were conducted to identify the structure-function relationship. Key residues on

asymmetric binding site (Met-26, Arg-37, Trp-38, His-44, Trp-142, Glu-144, Arg-152) or symmetric binding site (Gln-96, Glu-97, Asn-108, Arg-120, Phe-121, Lys-135, Arg-139, Glu-143) were mutated to alanine, respectively. By co-immunoprecipitation experiments, we found that mutation of asymmetric binding site significantly compromised the self-association of BcSEFIR, while it had minimum impact on the BcSEFIR $_{\Delta tail}$ symmetric interaction. On the other hand, mutation of symmetric binding site greatly reduced the dimerization of BcSEFIR $_{\Delta tail}$, but with little impact on self-association of BcSEFIR (Fig. 6A and B). These results provide supporting evidence for the two interaction interfaces revealed by our crystal structures, which is consistent with the proposed two modes of interaction (asymmetrical and symmetrical). Additionally, the mutagenesis experiments also suggest that with the tail region, SEFIR domain will prefer to form dimer through the asymmetric interface, as without the tail region, symmetric interaction

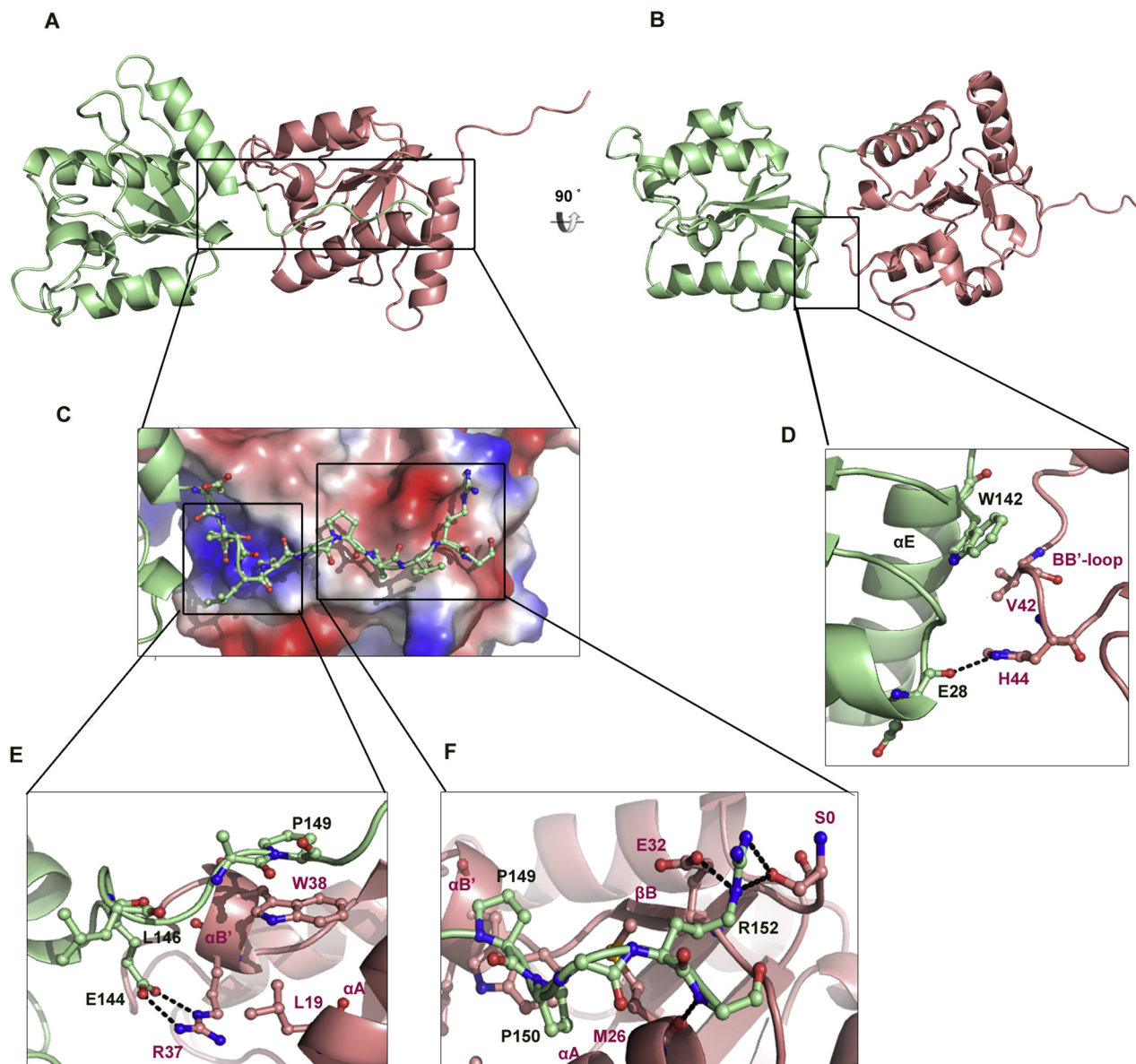


Fig. 3. The interface formed by the protruding tail of BcSEFIR. (A) In this asymmetric dimer of BcSEFIR, the tail-presenting molecule is colored in light green, as well as the tail-docking one in light pink. Helices αA , αB , $\alpha B'$ are involved in the interaction. Among them, αB , $\alpha B'$ play the most important role. Interface formed by the tail portion is highlighted with boxes. (B) The asymmetric dimer is shown in different orientation, interface is highlighted with a box. (C) The zoomed-in view of the protruding tail attaching to the adjacent BcSEFIR molecule. (D) Detailed interactions of the αE and αA helices in the tail-presenting BcSEFIR with the αB in the tail-docking BcSEFIR. (E) Detailed interactions of the N-terminal tail and intermediate portion in the tail-presenting BcSEFIR with the αA , $\alpha B'$ helices in the tail-docking BcSEFIR. Interacting residues are labeled, hydrogen-bonds are shown as dashed lines. (F) Detailed interactions of the tail C-terminal portion in the tail-presenting BcSEFIR with the αA , $\alpha B'$ helices and βB in the tail-docking BcSEFIR, where interacting residues are labeled, and hydrogen-bonds are shown as dashed lines.

is the dominant interaction mode.

3. Discussion

Both SEFIR family and TIR family belong to the STIR superfamily, of which the members share similar α/β fold and are known to mediate physical protein-protein interactions in cytosolic signal transductions. Several TIR-TIR complex structures have been reported, including human, plant and microbial TIR domains (e.g. TLR2, TLR6, TLR10, MAL/TIRAP, IL-1RAPL, L6, RPS4, RRS1TcPB, PdTIR etc.) (Tao et al., 2002; Valkov et al., 2011; Lin et al., 2012; Nyman et al., 2008; Alaidarous et al., 2014; Jang and Park, 2014; Chan et al., 2010; Bernoux et al., 2011; Snyder et al., 2014; Khan et al., 2004). Structural analysis combined with biochemical data revealed that TIR domain forms various different kinds of dimers. For example, the homodimer of

TLR10-TIR adopts a symmetric association mode, where the interactions at the homodimeric interface are mainly contributed by BB-loop and αC helix from both sides (Fig. S7D) (Nyman et al., 2008). The C713S mutant of TLR2-TIR was used for crystallization and unraveled an asymmetric homodimerization interface, where an inter-molecular disulfide bond is formed between the Cys-750 of one subunit and the Cys-640 of the other one (Fig. S7A) (Tao et al., 2002). In bacterial TcpB-TIR, the structural elements involved in these interfaces include the DD and EE loops and the αD and αE helices (Fig. S7F) (Snyder et al., 2014). In this study, we determined the crystal structures of BcSEFIR from a prokaryotic species, *Bacillus cereus* F65185, and revealed two novel SEFIR-SEFIR interaction modes (asymmetrical and symmetrical) mediated by two distinct interaction interfaces on SEFIR domains, which are totally different from any previously reported TIR-TIR complex structures. We further validate these two binding sites through

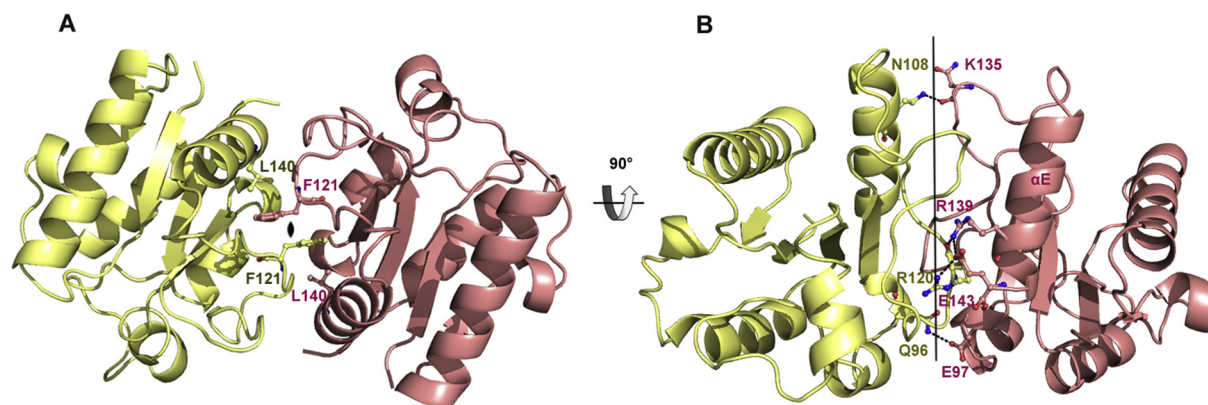


Fig. 4. Symmetric dimer of BcSEFIR Δ tail. (A) Two SEFIR domains interact with each other symmetrically in the structure of BcSEFIR Δ tail, residues which form a central hydrophobic core are labeled. (B) The symmetric dimer is shown in a different orientation.

site-directed mutagenesis. Our data showed that mutation of asymmetric binding site significantly compromised the self-association of BcSEFIR without substantial impact on BcSEFIR Δ tail symmetric interaction. Meanwhile, mutation of symmetric binding site impaired the self-association of BcSEFIR Δ tail, but not the dimerization of BcSEFIR (Fig. 6A and B), suggesting the two interaction interfaces on SEFIR domain are physiologically relevant to BcSEFIR domain self-association. We performed SPR experiments to quantitatively investigate the BcSEFIR-BcSEFIR interaction in solution, found that BcSEFIR domain showed weak affinity for self-association in solution (Fig. S5). The SPR results were consistent with the SEFIR-SEFIR interaction modes we observed in crystals, that BcSEFIR domain has different binding sites, and the tail region does not disturb the binding of the other interface. Jointly, our structural and biochemical data demonstrate two novel SEFIR-SEFIR interaction modes.

The asymmetric interaction mode implicates that SEFIR domains can create a fiber-like oligomeric structure as observed in the crystals (Fig. S4). We cannot rule out the possibility that this configuration might be the result of crystal packing in the high protein concentration

during crystallization. Notably, SEFIR domains appear to have the ability to form oligomers larger than dimer *in vivo*, as well as *in vitro* yeast two-hybrid and co-immunoprecipitation experiments of Act1 SEFIR domain, which is, in fact, critical for Act1-mediated downstream signaling (Mauro et al., 2003). Recently, Ve et al. reported that TIR domains of the TLR adaptors MAL and MyD88 can self-assemble into filaments, consisting of two parallel strands of TIR-domain subunits in a head-to-tail arrangement with two major types of asymmetric interaction (Ve et al., 2017). Although the assemblies mediated by Act1 SEFIR remain structurally elusive; previous studies have shown that the helix α B' and α B in Act1 SEFIR domain are essential for homotypic SEFIR-SEFIR interactions, since deletion part of the helix α B' and α B (residues 425–432) in Act1 SEFIR domain greatly reduced the self-association without significant impact on the heterotypic interaction of Act1 with IL-17RA-SEFIR (Zhang et al., 2013; Liu et al., 2011). In this study, our crystal structures indicate that helix α B' and α B are part of the asymmetric interaction interface for creation of fiber-like oligomeric structure. One thing to be noted is that the helix α B' in all three SEFIR structures is locked into a rigid conformation by hydrophobic

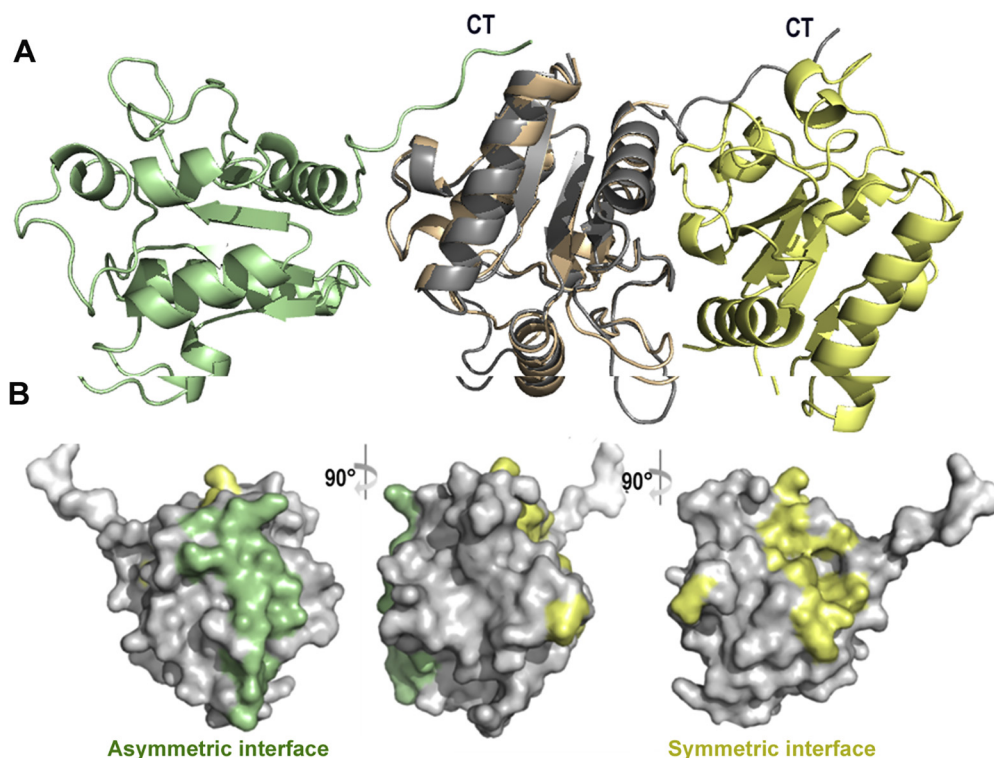


Fig. 5. Two distinct SEFIR-SEFIR interfaces. (A) Two distinct SEFIR-SEFIR interaction modes are aligned together, where asymmetric dimer are colored in green and grey, while symmetric dimer are coloured in light orange and yellow. C-terminal tail regions are labeled as CT. (B) The asymmetric (green) and symmetric interfaces (yellow) are located at different regions on the surface of BcSEFIR domain.

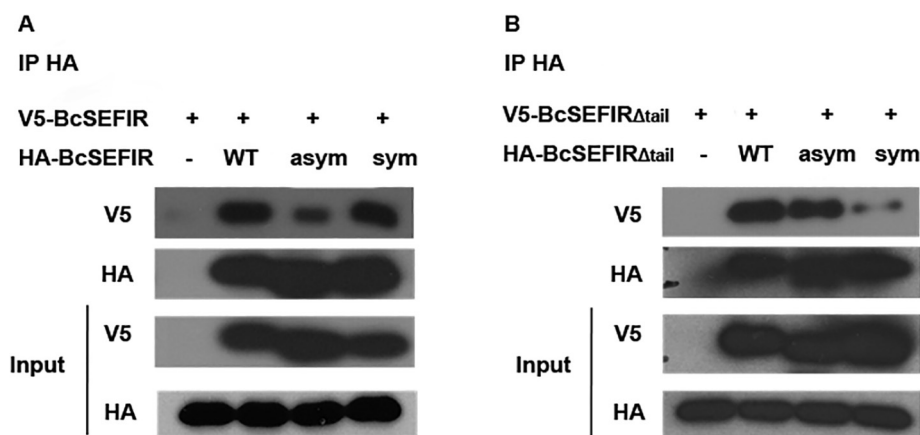


Fig. 6. Co-immunoprecipitation assays. (A) The ability of V5-BcSEFIR to co-immunoprecipitate with the HA-mutations or wild type HA-BcSEFIR was shown. Mutagenesis on asymmetric interface markedly reduced BcSEFIR self-association. (B) The ability of V5-BcSEFIR Δ tail to co-immunoprecipitate with the HA-mutations or wild type HA-BcSEFIR Δ tail was shown. Mutagenesis on symmetric interface markedly reduced BcSEFIR Δ tail self-association. WT, wild type recombinant protein. Asym, mutant with key interacting amino acids on asymmetric binding site substituted by alanine. Sym, mutant with key interacting amino acids on symmetric binding site substituted by alanine.

interactions and hydrogen bond (Zhang et al., 2013, 2014). The helix α B' in IL-17RA-, IL-17RB forms hydrogen bond with helix α B, the related residues are conserved among IL-17Rs-SEFIR and Act1-SEFIR (Fig. S2, green boxes). While the helix α B' in BcSEFIR forms hydrogen bond with AA-loop, the related residues are also strictly conserved among IL-17Rs-SEFIR and Act1-SEFIR (Fig. S2C, red boxes). The residues involved in hydrophobic interactions are also conserved, indicating that α B' in Act1-SEFIR is tightly tethered. The stable conformation of helix α B' suggested that α B' may play important roles during SEFIR domain assembly.

According to the multiple sequence alignment, the BcSEFIR tail region is highly conserved among Act1 SEFIR domains, including three absolutely conserved residues that constitute an EXXXXPP motif (Fig. S3). The highly conserved two adjacent proline residues form the core of the hydrophobic interactions on the tail. As for the binding surface of the tail-docking molecule, Asn-23 and Trp-38 are the most critical hydrophobic residues interacting with the proline residues on the tail region, which are conserved in Act1-SEFIRs (Fig. S3, showed in green boxes). Conserved hydrophobic residues Leu-19, Gly-30, Val-31 are also involved in the interaction with tail region. The highly conserved glutamate residue in EXXXXPP motif is the N-terminal anchoring point of the tail in asymmetric interaction mode, it forms two hydrogen-bonds and one ionic bond with Arg-37 in BcSEFIR, but in Act1-SEFIR this arginine residue is substituted by isoleucine or valine, which cannot form hydrogen-bonds (Fig. S3, showed in blue boxes). In the symmetrical interaction mode, interacting residues on helix α E are conserved and constitute an RLI/LXE motif near the C-terminus of Act1 SEFIR domains. The absolutely conserved residues Arg-139, Leu-140, Glu143 interact with the Arg-120 from the opposite molecule, form three hydrogen bonds and one salt bridge (Fig. 3B). In Act1-SEFIR, Arg-120 is substituted by threonine (Fig. 1A), with adjacent residues of asparagine and histidine, where hydrogen-bonds might also be formed.

In summary, our study reveals that there are two distinct SEFIR-SEFIR interfaces (asymmetrical and symmetrical) exist on BcSEFIR domain, illustrates two plausible modes for SEFIR-mediated protein-protein interactions. However, future studies are required to determine whether these interaction modes are shared by Act1-SEFIR in IL-17 signaling.

4. Materials and methods

4.1. Cloning, expression, and purification

The two coding sequences of BcSEFIR Δ tail (residues 1-143) and BcSEFIR (residues 1-153) were cloned into a modified pET28a vector which have a fused SUMO protein with 6 $\times$ His tag at N terminal. The cells were grown at 37 °C to an optical density of 0.6 and then induced with 0.2 mM isopropyl-D-thiogalactopyranoside at 16 °C overnight.

Bacterial pellets were then resuspended in lysis buffer (20 mM Tris- HCl pH 8.0, 150 mM NaCl) and lysed by sonication at 4 °C. The fusion proteins were purified by Ni-affinity chromatography, followed by digestion of protease Ulp1 to remove N terminal 6 $\times$ His tag and fused sumo, then proteins were purified by anion exchange chromatograph for further investigations. Selenomethionine (SeMet)- substituted BcSEFIR was expressed in M9 minimal medium supplemented with amino acids (Van Duyne et al., 1993) and purified similarly to the native protein using the procedures described above. Mutations DNA sequences were synthesized commercially. For the immunoprecipitation experiments, the BcSEFIR or mutations DNA sequences with a N-terminal HA-tag or V5-tag were generated by PCR methods. The new DNA inserts were subsequently subcloned into the plasmid described before. The HA-tagged or V5-tagged proteins were expressed and purified as native proteins.

4.2. Crystallization

The BcSEFIR Δ tail and BcSEFIR were both crystallized at 16 °C using the hanging-drop vapor-diffusion method by equilibrating a mixture containing 1 μ l of protein solution (8 mg/ml in a buffer of 20 mM Tris-HCl pH 8.0, 150 mM NaCl) and 1 μ l of reservoir solution (0.2 M NaCl, 0.1 M imidazole pH 8.0, 30% PEG 8000) against 1 ml of reservoir solution. After one week, single crystals formed and were flash-frozen in liquid nitrogen for future data collection. The crystal of selenomethionine-labeled BcSEFIR grew in a different reservoir solution (0.2 M Lithium sulfate monohydrate, 0.1 M Bis-Tris pH 5.5, 25% PEG 3350).

4.3. Data collection, phasing and model refinement

Diffraction data for BcSEFIR crystals were collected at beamline BL19U1 of Shanghai Synchrotron Radiation Facility. Diffraction data for BcSEFIR Δ tail crystals were collected using Rigaku FR-E/VariMax HR/Raxis IV ++ (Rigaku, Japan) at the Structural Biology Core Facility (Institute of Biophysics, Chinese Academy of Sciences). Data were indexed and scaled with HKL2000 (Otwinowski and Minor, 1997). Phasing problems of BcSEFIR was solved by using multiwavelength anomalous diffraction method (Kahn et al., 1985), and there are three selenium sites in each asymmetric unit in the structure. Then the structure of BcSEFIR Δ tail was solved by using molecular replacement method performed on PHENIX.phaser (Zwart et al., 2008), with the structure of BcSEFIR as a search model. Refinement was carried out with PHENIX.refine (Zwart et al., 2008) and COOT (Emsley and Cowtan, 2004). All the statistics of data collection and refinement are listed in Table 1. The completeness in the high resolution shell for BcSEFIR Δ tail is relatively low. Coordinates and structure factors have been deposited in the Protein Data Bank with accession numbers 5Y8F and 5Y8E for structures of BcSEFIR Δ tail and BcSEFIR, respectively.

Table 1

Data collection and refinement statistics.

	BcSEFIR	BcSEFIR _{tail}
Crystallographic data		
d _{min} (Å)	1.80 (1.80–1.83)	2.00 (2.00–2.05)
Wavelength (Å)	0.9793	1.5418
Measured reflections	108121 (3104)	271348 (1654)
Average redundancy	4.1 (4.1)	2.8 (2.0)
Mean I/σ (I)	11.2 (8.6)	12.7 (2.2)
Completeness (%)	99.4 (99.2)	94.9 (67.3)
R _{merge} ^a	0.097 (0.218)	0.079 (0.391)
Refinement statistics		
Bragg spacing (Å)	35.33–1.80	29.18–2.00
Space group	P2 ₁	P2 ₁
Cell parameters		
a (Å)	38.74	37.86
b (Å)	61.67	68.02
c (Å)	39.27	54.94
Reflections in working set	14504	16937
Reflections in test set	765	917
R _{cryst} ^b	0.1668	0.1857
R _{free} ^c	0.1963	0.2219
r.m.s.d. bonds (Å)	0.006	0.004
r.m.s.d. angles (°)	0.963	0.982
Average B-factor (Å ²)	22.48	25.38
No. of residues	154	288
No. of waters	223	346

Values in parentheses indicate the corresponding statistics in the highest resolution shell.

^a $R_{\text{merge}} = (I_i - d_i)/I_i$, where I_i is the integrated intensity of a given reflection.

^b $R_{\text{cryst}} = (|F_o| - |F_c|)/|F_o|$, where F_o and F_c denote observed and calculated structure factors, respectively.

^c R_{free} is equivalent to R_{cryst} but calculated using randomly chosen 5% reflections as the test set, which were excluded from the refinement process.

4.4. Immunoprecipitation

For co-immunoprecipitation assays to validate the structure-function relationship, 0.5 μg purified V5-tagged BcSEFIR or BcSEFIR_{tail} were mixed 1:1 with HA-tagged mutations or wild type BcSEFIRs for 30 min in 4 °C in immunoprecipitation (IP) buffer (20 mM Tris-HCl pH 8.0, 150 mM NaCl). Protein G dynabeads were added to each sample, then incubated with anti-HA antibody (Cell Signalling Technology) at 4 °C overnight, beads were washed four times in IP buffer, boiled in Laemmli sample buffer, and analyzed by SDS-PAGE/immunoblotting. Blots were labeled using anti-V5 and anti-HA antibodies (Invitrogen), while detection was carried out using ECL reagents (GE Healthcare).

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.jsb.2018.03.005>.

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